Voice Assistant with Emotion Recognition*

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***Abstract*—This paper presents an AI voice assistant capable of detecting and responding to human emotions using a deep learning-based facial emotion recognition system. Inspired by the YOLO (You Only Look Once) framework, our approach enables real-time emotion analysis, facilitating context-aware interactions. This system has potential applications in psychological support, customer service, and personalized AI-driven experiences.**

I. INTRODUCTION

Recent advances in artificial intelligence (AI) have paved the way for more sophisticated human-computer interactions. Emotion recognition plays a critical role in understanding user needs and enhancing communication. Facial Emotion Recognition (FER) refers to the process of identifying and categorizing human emotions based on facial expressions. By analyzing facial features and patterns, machines can learn about a person's emotional state. This subfield of facial recognition is highly interdisciplinary, drawing on insights from computer vision, machine learning, and psychology [1]. This project aims to develop a voice assistant that can detect and analyze users' emotions through facial expression recognition and provide relevant responses. By combining deep learning-based emotion detection with NLP-based dialogue systems, this assistant can serve as an interactive and empathetic AI companion.

II. OBJECTIVES

- Develop a deep learning model for real-time facial emotion recognition with opencv, trying different CNN structures.
- Construct a labeled dataset containing facial images annotated with emotional states.
- Implement a voice-based assistant capable of processing and responding to spoken language.
- Integrate emotional analysis with NLP techniques to generate context-aware responses.
- Explore potential applications in psychological support and mental well-being.

III. METHODOLOGY

- Dataset Collection: Acquire and preprocess a dataset of facial images labeled with emotions such as happiness, sadness, anger, and surprise.
- Deep learning model: Develop different CNNs and demonstrate which one is the best.
- Speech Processing: Implement NLP techniques for speech recognition and response generation.
- System Integration: Combine vision and NLP components to enable the assistant to detect emotions and provide context-aware responses.
- Evaluation: Assess system performance using metrics such as f1-score, response relevance and user satisfaction.

IV. FACIAL EMOTION RECOGNITION

A. VGG-like with FER2013

1) Introduction: The goal is to develop a deep learning model that can accurately classify facial expressions into one of seven categories: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. I try a CNN model with a VGG-like structure to train the model with the FER2013 dataset.

2) Data: The FER-2013 dataset consists of 48x48 pixel grayscale images of faces that have been automatically registered to be centred and occupy a similar amount of space in each image. The dataset contains 24,400 images, with 22,968 examples in the training set and 1,432 examples in the public test set.

3) Preprocessing: I used data generators and data augmentation to train the deep learning model:

- Data generators create batches of images on-the-fly during training, making it possible to handle this large and complex dataset efficiently. Since loading all images into memory at once would be impractical, data generators allow the model to be trained on the entire dataset without exceeding memory limits.
- Data augmentation techniques to the training images, such as rotation, shifting, and flipping for increasing the size and diversity of the training set, which can improve the performance of the deep learning model. By introducing variability into the training process, data augmentation can also help to prevent overfitting, which occurs when the model becomes too closely tailored to the training set and performs poorly on new, unseen data.

4) Training: To develop the CNN, at first I defined the the input layer and the number of filters in the first convolutional layer. Then, I added additional convolutional layers with increasing numbers of filters, followed by max-pooling layers to reduce the spatial dimensions of the feature maps. After the convolutional layers, I added fully connected layers with ReLU activation to classify the emotions.[2]

I experimented with different numbers of convolutional layers, filter sizes, and fully connected layers to optimize the model performance. Additionally, I used techniques such as dropout and batch normalization to prevent overfitting and improve the generalization capability of the model.

The structure that I finally chose is:

TABLE I
SUMMARY OF THE CNN ARCHITECTURE

Layer	Filters	Kernel Size	Activation
Input	-	$48 \times 48 \times 1$	-
Conv2D + BatchNorm	32	3×3	ReLU
Conv2D + BatchNorm	64	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Conv2D + BatchNorm	128	3×3	ReLU
Conv2D + BatchNorm	128	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Conv2D + BatchNorm	256	3×3	ReLU
Conv2D + BatchNorm	256	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Flatten	-	-	-
Dense + BatchNorm	256	-	ReLU
Dropout	-	-	0.5
Dense (Output)	7	-	Softmax

5) Evaluation: The training and validation loss function decreased over epochs with a final loss in the epoch 50 of loss: 0.9036 that it typically means the model is not performing optimally.

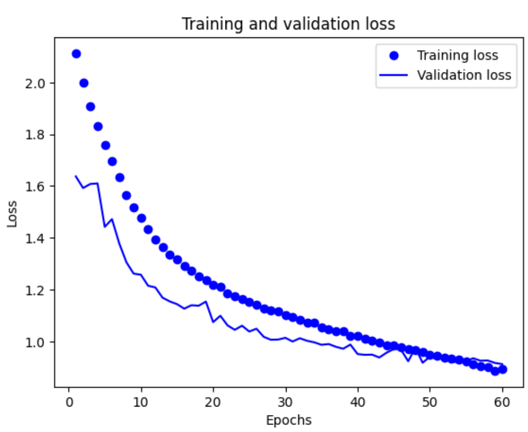

Fig. 1. Training and validation loss graph.

The training and validation accuracy increased over epochs with a final accuracy of 0.6631 that is not enough.

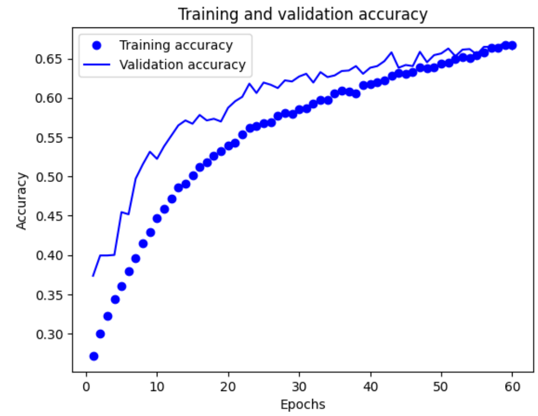

Fig. 2. Training and validation accuracy graph.

In the confusion matrix there are a lot of false positives and false negatives showing not enough accuracy. Besides, there are 0 disgust labeled images in the validation, this shows an imbalance in the data.

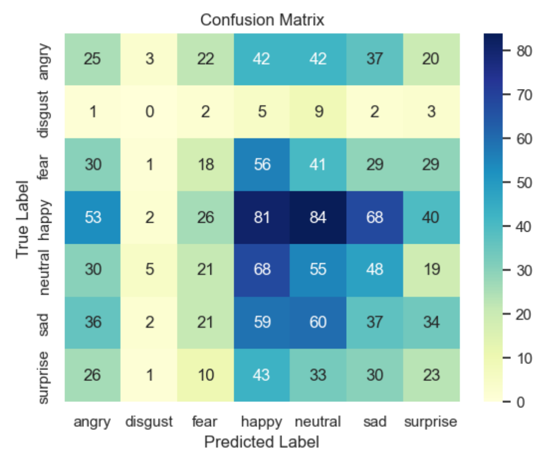

Fig. 3. Confusion matrix.

6) Testing: I used the haarcascade_frontalface_default.xml classifier from OpenCV, that is a widely used pre-trained model for detecting faces in images. Ergo, I recognized the faces with the haarcascade classifier and later recognise the facial emotion with the previous CNN. The haarcascade uses Haar features and is based on the Viola-Jones algorithm, which was one of the first methods to offer a real-time face detection solution. While it is quite efficient and works well for many applications, its performance can be limited by a complex environment, faces not in a frontal position and has lower accuracy compared to modern methods.

On the test that I did, the performance of the haarcascade was regular predicting sometimes false positives (it predicted as faces things that aren't a face) and false negatives above all when the face was not frontal. The performance of the VGG-like for facial emotion recognition wasn't good enough predicting the most of times neutral and happy when the true label were others.

B. *emotion-ferplus-8*

1) Data: I used the FER+ dataset is a notable extension of the original Facial Expression Recognition (FER) dataset (the one that I used previously). Developed to improve upon the limitations of the original dataset, FER+ offers a more refined and nuanced labeling of facial expressions. The FER+ has 8 emotions: happiness, sadness, anger, surprise, fear, disgust, neutral and contempt.

2) RFB-320 Single Shot Multibox Detector (SSD) Model for Face Detection: Before the emotion is recognized, the face needs to be detected in the input frame like in the previous case. For this, the Ultra-lightweight Face Detection RFB-320 is used. It is a face detection model optimized for edge computing devices. Incorporating a modified Receptive Field Block (RFB) module (an extension of traditional convolutional layers), it effectively captures multiscale contextual information without adding computational overhead. Trained on the diverse WIDER FACE dataset (32,203 images and labeled 393,703 faces with a high degree of variability in scale, pose, and occlusion) and tailored for 320×240 input resolution, it boasts an impressive balance of efficiency with a computational rate of 0.2106 GFLOPs and a compact size housing of 0.3004 million parameters. Developed in the PyTorch framework, the model achieves a commendable mean average precision (mAP) of 84.78%, marking its position as a potent solution for efficient face detection in resource-constrained environments. In this implementation, the RFB-320 SSD model has been used in Caffe format (the data storage and model representation format used by Caffe, a deep learning framework). [1]

3) Custom VGG13 Model Architecture: The emotion recognition classification model employs a customized VGG13 architecture designed for 64×64 grayscale images. It classifies images into eight emotion classes using convolutional layers with max pooling and dropout to prevent overfitting. This enhances the model's ability to generalize and accurately recognize emotions in various images. It was training using the FER+ dataset. The structure is:

TABLE II
CUSTOMIZED VGG13 MODEL ARCHITECTURE FOR FACIAL EMOTION RECOGNITION

Layer	**Filters**	**Kernel Size**	**Activation**
Input	-	$64 \times 64 \times 1$	-
Conv2D	64	3×3	ReLU
Conv2D	64	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	128	3×3	ReLU
Conv2D	128	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Fully Connected (Dense)	1024	-	ReLU
Dropout	-	-	0.50
Fully Connected (Dense)	1024	-	ReLU
Dropout	-	-	0.50
Fully Connected (Dense)	8	-	ReLU
Softmax (Output)	8	-	Softmax

4) Implementation: At first, I initialize these parametres:

- image_mean: Mean values for image normalization across RGB channels.
- image_std: Standard deviation for image normalization.
- iou_threshold: Threshold for Intersection over Union (IoU) metric to determine bounding box matches. The best prediction is the one that has the highest intersection with the ground truth while minimizing the area outside it.
- center_variance: Scaling factor for predicted bounding box center coordinates.
- size_variance: Scaling factor for predicted bounding box dimensions.
- min_boxes: Minimum bounding box dimensions for objects of different sizes.
- strides: Control the scale of the feature maps according to the image size.
- threshold: Confidence threshold for object detection.

5) Testing: The ultralightweight face detection RFB320 is very good and accurate recognizing the faces although the quantity, angles of the face, illumination... The custom VGG13 model architecture performed well in classifying emotions from detected faces. It excelled at recognizing happiness, neutrality, and surprise, while showing slightly lower accuracy for sadness, disgust, and fear

REFERENCES

[1] LearnOpenCV, "Facial emotion recognition," 2023, accessed: March 4, 2025. [Online]. Available: https://learnopencv.com/facial-emotion-recognition/

[2] M. Chahed, “Human emotion detection,” 2023, accessed: 2025-03-22. [Online]. Available: https://www.kaggle.com/code/mohamedchahed/human-emotion-detection/notebook